

Evaluating Adjustment and Proficiency Disparities in Virtual Reality

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ABSTRACT

Our study explores the relationship between VR experience, 3D gaming expertise, and performance metrics like task completion time, accuracy, and workload. Analysis reveals that combined expertise in VR and 3D gaming improves efficiency, reduces workload, and enhances accuracy. Notably, users with experiences in both domains consistently perform better, with higher accuracy and lower task-load scores, demonstrating the value of VR proficiency. Interestingly, physical space requirements remain low across all user levels, highlighting VR's accessibility.

Index Terms: Human Computer Interaction, Virtual Reality, NASA-TLX

1 INTRODUCTION

Virtual Reality (VR) is increasingly used for precision-based tasks such as virtual surgery, remote-controlled robotics, and industrial design [3, 5, 2]. These tasks require fine motor control and spatial awareness, skills often influenced by prior VR and 3D gaming experience. Video gaming enhances hand-eye coordination, spatial awareness, and decision-making [5, 4], making it an important factor in effectively navigating VR environments.

Our study investigates the combined impact of VR experience and 3D gaming expertise on performance in precision-based tasks using a VR adaptation of the buzz-wire game (cf. Fig. 1) where participants guide a loop along a virtual wire without making contact only relying on visual feedback and precise hand movements. Our hypothesis posits that, while VR and 3D gaming experience individually contribute to performance, combined expertise in both domains offers superior accuracy, spatial efficiency, and reduced task load.

Examining these relationships, our research fills a gap in understanding how multidimensional expertise influences performance in VR, providing insights to optimize training and task design in immersive environments.

2 METHODOLOGY

Our study investigated the influence of VR familiarity and video gaming expertise on performance in precision-based tasks using an adaptation of the traditional buzz-wire game in a VR environment. The VR application was designed to assess precision, spatial awareness, and motor control.

2.1 Task Design

The VR application required participants to guide a loop along a virtual wire without making contact. Visual feedback was integrated into the game to guide participants and provide real-time error correction (cf. Fig. 1). The game started by presenting par-

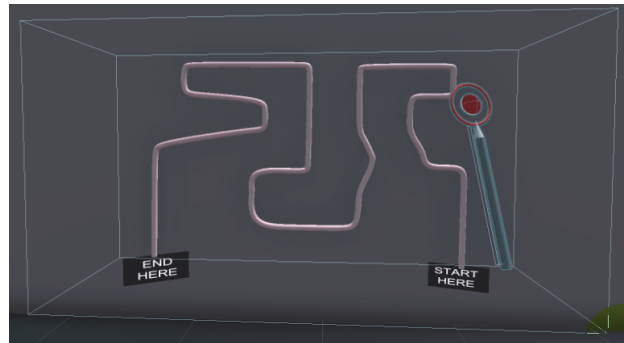


Figure 1: Virtual buzz-wire game setup. Participants guide a virtual loop along a wire using their chosen hand. The loop and its inner red circle turn black upon wire contact, offering real-time positioning feedback. The collider boundary (green lines) defines the designated *playing zone* in the VR task.

ticipants with a scene to familiarize themselves with the VR environment and its controls. Participants approached a virtual table with two buttons, one for a left-hand setup and the other for a right-hand setup. Hovering over a button selected their preferred hand configuration.

Once the task started, participants navigated the loop through the wire while avoiding contact. Upon touching the wire, the loop and an inner red circle changed color to black, providing immediate visual feedback and prompting participants to adjust their movements. The task concluded when participants hovered their hand over an *end task* button, transitioning to a final scene displaying an award mural to signify completion. This design allowed precise tracking of task-performance times while maintaining participant engagement.

2.2 Hypothesis

Our study employed a hypothesis-driven approach:

- **Null Hypothesis (H_0):** Neither VR experience nor 3D gaming expertise, individually or in combination, significantly affect performance in precision-based VR tasks.
- **Alternative Hypothesis (H_1):** Combined proficiency in both VR and 3D gaming expertise enhances performance in precision-based VR tasks.

Participants ($n = 35$) with diverse demographic backgrounds and varying levels of VR and video gaming expertise were recruited. Participants were asked to complete a pre-experience questionnaire, a post-experience survey, and the Simulator Sickness Questionnaire (SSQ) to capture baseline data and assess the impact of the VR task on their performance and comfort.

2.3 Hardware and Simulation

Our study utilized a Meta Quest 3 VR headset and hand-held controllers for immersive interaction. The VR environment was developed using the Unity game engine with an OpenXR integration. Custom scripts enabled seamless scene management, task progression, and high-frequency data capture at 10 ms intervals. Key performance metrics such as task-completion time, collision events, hand and head movements, and spatial engagement were recorded for detailed offline analysis.

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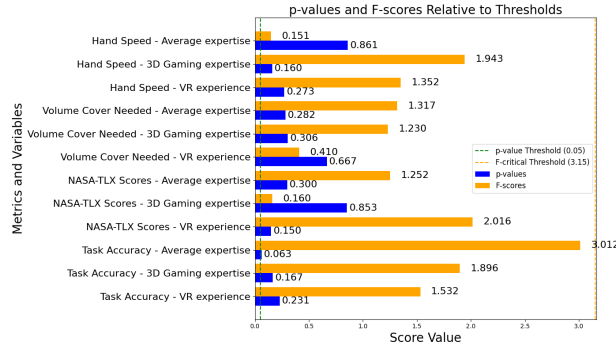


Figure 2: Combined visualization of p-values and F-scores for VR experience, 3D gaming expertise, and average expertise across key metrics, highlighting their significance relative to the thresholds (p-value: 0.05, F-critical: 3.15).

2.4 Data Collection and Analysis

The captured data included collision counts, spatial movement volumes, hand speeds, and the duration of task engagement. All data were anonymized to ensure participant confidentiality. The analysis focused on identifying the impact of VR and gaming expertise, as well as their combined influence, on task accuracy, workload, and spatial efficiency. Our study underwent IRB approval at the hosting institution to ensure ethical compliance.

3 RESULTS

A Pearson correlation matrix [1] was used to evaluate the relationship between VR experience, 3D gaming expertise, and combined expertise in both domains with key performance metrics. Combined expertise showed the strongest positive correlation with task accuracy ($r = 0.366$) and moderate positive correlations with 3D space utilization ($r = 0.151$). Additionally, it was negatively correlated with task load ($r = -0.26$), indicating reduced perceived workload for participants with higher combined expertise.

Analysis of variance [1] (ANOVA) was utilized to evaluate the influence of VR experience, 3D gaming expertise, and combined expertise in both domains on key performance metrics. Participants were categorized into three percentile groups, i.e. high (80–100%), medium (30–80%), and low (0–30%), based on their perceived levels of VR and 3D gaming expertise. Despite the limited sample size ($n = 35$) the grouping allowed for a detailed examination of statistical trends. The study employed a 2×3 factorial design with two independent variables: *VR Experience* (Low, Medium, High) and *Gaming Expertise* (Low, Medium, High). The dependent variables analyzed included *Accuracy*, *Task Load* (NASA-TLX), *Task Completion Time*, *Spatial Efficiency*, and *Hand Speed* to assess both individual and interaction effects across expertise levels.

Our Key findings are as follows:

Task Accuracy: As shown in Fig. 2, combined expertise ($F = 3.012, p = 0.063$) showed a stronger potential influence, nearing significance, compared to individual VR experience ($F = 1.532, p = 0.231$) or 3D gaming expertise ($F = 1.896, p = 0.167$). Participants with combined expertise in both domains demonstrated higher and more consistent accuracy.

Task Load (NASA TLX): Neither VR ($F = 2.016, p = 0.150$) nor 3D gaming expertise ($F = 0.160, p = 0.853$) significantly reduced task load. Combined expertise ($F = 1.252, p = 0.300$) indicated a potential trend toward lower perceived workload (cf. Fig. 2). The grouping strategy helped to reveal subtle differences across expertise levels.

3D Space Utilization: No significant differences were found across expertise levels, though combined expertise in both domains ($F = 1.317, p = 0.282$) suggests more efficient spatial use compared to VR experience or 3D gaming expertise alone (cf. Fig. 2). The grouping approach highlighted nuanced trends in spatial efficiency.

Hand Speed: Hand speed showed no significant effect across VR experience ($F = 1.352, p = 0.273$), 3D gaming expertise ($F = 1.943, p = 0.160$), and combined expertise in both domains ($F = 0.151, p = 0.861$) as shown in Fig. 2. 3D gaming expertise indicated a potential trend toward faster movements. Small sample size may have limited statistical power in detecting significant effects.

4 CONCLUSIONS AND FUTURE WORK

The analysis of key performance metrics, including task load, accuracy, and volumetric area covered, reveals that combined expertise in VR and 3D gaming significantly enhances performance in precision-based spatial tasks. In contrast, focusing solely on VR experience or 3D gaming experience alone did not yield substantial improvements with ANOVA results for these individual factors failing to reach statistical significance ($p > 0.05$). The strongest correlations and most impactful F-scores were observed in participants with combined expertise in VR and 3D gaming, where p-values approached the threshold of significance. These results demonstrate that integrating skills from both domains yields better performance, supporting the alternative hypothesis.

Our findings highlight the importance of expertise in multiple domains for tasks requiring precise motor control. Combined expertise not only improves accuracy. It also contributes to reduced task-load scores as well as more efficient spatial usage. Participants with higher combined expertise exhibited better task management and adaptability, suggesting that integration of skills across domains leads to enhanced efficiency and reduced cognitive strain. These insights provide a basis for designing VR systems that emphasize adaptability and user-specific training for enhanced performance and usability.

Future research will have to expand the sample size to improve generalizability and statistical power of our experimental setup. This will allow to better explore how demographic factors, such as athletic skills, age, and physical attributes influence VR task performance. For example, attributes such as coordination, flexibility, and user height may significantly affect motor control and spatial awareness, which may help to refine the design of adaptive VR systems tailored to diverse groups of users.

Data limitations can be addressed by leveraging synthetic data-generation methods (e.g., Gaussian Mixture Models, Variational Autoencoders, Copulas) to enrich experiment datasets. Integrating synthetic insights with real-world data allows developing personalized VR applications adaptable to individual expertise levels and physical traits. This would pave the way for innovative, accessible, and user-focused VR applications.

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